

Novel Joint Transfer Network for Unsupervised Bearing Fault Diagnosis From Simulation Domain to Experimental Domain

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I. INTRODUCTION

Abstract—Unsupervised cross-domain fault diagnosis of bearings has practical significance; however, the existing studies still face some problems. For example, transfer diagnosis scenarios are limited to the experimental domain, cross-domain marginal distribution and conditional distribution are difficult to align simultaneously, and each source-domain sample is assigned with equal importance during the domain adaptation process. Aiming at the above-mentioned challenges, this article proposes a novel joint transfer network for unsupervised bearing fault diagnosis from the simulation domain to the experimental domain. The sufficient bearing simulation data containing rich fault label information are used to construct the source domain to reduce the dependence on the resources of laboratory test rigs. An improved loss function embedded with joint maximum mean discrepancy is designed to achieve simultaneous alignments of marginal and conditional distributions across domains in unsupervised scenarios. A weight allocation mechanism for each source-domain sample is developed to suppress negative transfer. Two experimental datasets collected from laboratory test rigs are used as the target domains to validate the effectiveness of the proposed method. The results show that the proposed method is superior to other popular unsupervised cross-domain fault diagnosis methods.

Index Terms—Improved loss function, novel joint transfer network (NJTN), simulation domain, unsupervised fault diagnosis, weight allocation mechanism.

ROTATING machinery plays an important role in the field of manufacturing industries [1]–[3]. Bearing is a common and key component of rotating machinery equipment, whose faults may cause economic loss and even human casualties. Therefore, bearing fault diagnostics is of great importance to the operational safety of mechanical equipment [4]–[8].

In recent years, thanks to the rapid development of artificial intelligence technology, deep learning models have been developed and successfully applied in the field of end-to-end machinery intelligent fault diagnosis, which have received extensive attention from scholars worldwide [9]–[16]. However, the excellent diagnostic performance of many deep learning models is inseparable from sufficient, labeled, and identically distributed samples. In practical engineering, it is difficult or even impossible to obtain many or even a few samples containing rich fault label information for reasons such as cost and safety. In addition, mechanical equipment often operates under variable conditions of load and rotational speed. As a result, the measured samples usually obey different distributions, so the deep learning models trained on samples under one distribution (source domain) are probably difficult to directly generalize to samples under another distribution (target domain) [17]–[21]. Therefore, research studies on unsupervised cross-domain fault diagnosis of bearings have more practical significance [22], [23].

Unsupervised domain adaptation is an important transfer learning strategy that can be used to solve the above-mentioned problems. In recent years, researchers have successively carried out research on unsupervised cross-domain fault diagnosis of bearings. In 2019, Yang *et al.* [24] detected various faults of bearings by reducing the distribution distance between domains through minimizing the maximum mean discrepancy (MMD). In 2019, a transfer learning fault diagnosis method for bearing based on multi-kernel MMD was introduced by An *et al.* [25]. In 2020, Jiao *et al.* [26] utilized an unsupervised adversarial adaptation network for intelligent fault diagnosis of different machines. In 2021, Zhao *et al.* [27] designed a multiscale adversarial transfer network with attention, which automatically distinguished the various fault states of bearings. In 2021, a domain adversarial network embedded with correlation alignment (CORAL) was proposed by Qin *et al.* [28] to identify the failure modes of bearings. In 2021, Li *et al.* [29] adopted an optimal ensemble deep transfer network to achieve automatic

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fault classification of bearings. In 2022, Liu *et al.* [30] developed a deep adversarial subdomain adaptation network to align the relevant distributions of subdomains of the same fault categories in the source domain and the target domain.

These above-mentioned methods have shown good transfer learning ability in various unsupervised cross-domain fault diagnosis tasks of bearing. However, some challenging problems still need to be solved to further the diagnosis performance.

- 1) Most of the source-domain fault data of bearing in the current research come from the laboratory test rigs, which are more appropriate for investigating the general law of failure phenomena. However, the construction of laboratory test rigs with a certain accuracy requires a large amount of long-term resource investment, which makes it difficult to flexibly meet the requirements of fault data under different working conditions. Fortunately, with the help of numerical simulation technology, a large number of simulation data with rich fault label information can be easily obtained, which can also reduce the resource dependence on the laboratory test rigs [31], [32].
- 2) The above-mentioned studies usually only consider the marginal distribution alignment between the source and target domains while ignoring the conditional distribution alignment, which may cause fuzzy samples close to the class boundary or far from the class center in the target domain to be incorrectly classified [33], [34]. Since there are no labeled samples in the target domain of unsupervised cross-domain fault diagnosis research, modeling its conditional distribution is extremely challenging. Therefore, it is urgent to design new loss functions that can achieve simultaneous alignments of conditional distribution and marginal distribution in unsupervised scenarios.
- 3) In the current research, each source-domain sample is usually assigned equal weight when performing domain adaptation. Even those source-domain samples that are quite different from the target domain are still given the same importance, which may cause negative transfer [35]. Consequently, it is necessary to develop a weight allocation mechanism for samples in the source domain, which is expected to adaptively assign different weights for each source-domain sample to suppress negative transfer and improve the diagnostic accuracy to a certain extent.

In this article, a new method called novel joint transfer network (NJTn) is proposed to diagnose various faults of bearing in unsupervised cross-domain scenarios automatically. Compared with various methods in two cases from the simulation domain to the experimental domain, the results demonstrate that the proposed method can achieve the best diagnosis performance. The innovations and contributions of the proposed method are summarized as follows:

- 1) Different from the existing transfer diagnosis research studies between experimental data, this article constructs the source domain by using simulation data with rich fault label information, reducing the dependence on the laboratory test rigs and exploring data–physical coupling driven fault diagnosis way.

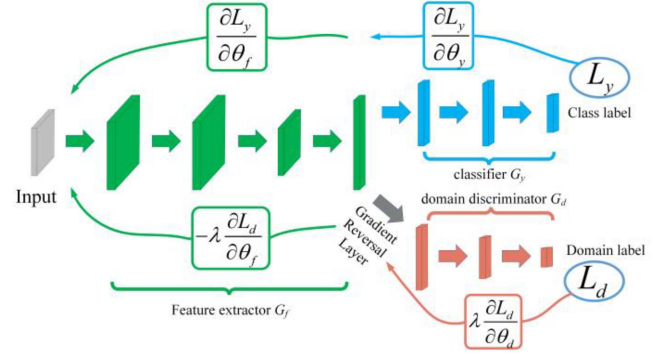


Fig. 1. Classic DANN model diagram.

- 2) An improved loss function embedded with joint MMD (JMMD) is designed to realize the simultaneous alignments of marginal distribution and conditional distribution in the unsupervised scenario.
- 3) A weight allocation mechanism is developed based on domain similarity, which assigns different weights to each source-domain sample to suppress negative transfer.

The rest of this article is organized as follows. Section II introduces the classical theory of unsupervised domain adversarial neural network (DANN). Section III elaborates on the proposed method. Section IV describes two cases of unsupervised cross-domain fault diagnosis of bearings. Finally, Section V concludes this article.

II. UNSUPERVISED DANN

DANN [36] is mainly used to solve the problem of unsupervised cross-domain data distribution mismatch, and it reduces the distribution discrepancy between the source domain and the target domain by adapting their marginal distributions. The structure of DANN is shown in Fig. 1, and it is mainly composed of feature extractor G_f , domain discriminator G_d , and classifier G_y , in which G_f is used to learn domain-invariant and discriminative feature representations; G_d tries to discriminate whether the feature representations transformed by G_f are from the source domain or the target domain; G_y is responsible for classification. In particular, G_f and G_d are connected by the gradient reversal layer.

Let $\mathcal{D}_s = \{x_i^s, y_i^s\}_{i=1}^{n_s}$ be the source domain with n_s labeled samples, and $\mathcal{D}_t = \{x_i^t\}_{i=1}^{n_t}$ be the target domain with n_t unlabeled samples. In this article, it is assumed that the source domain and the target domain share the same label space, which means that they have the same health state categories. The common loss function of DANN consists of categorical cross-entropy loss L_y and domain discriminative cross-entropy loss L_d , and it can be defined as follows:

$$L(\theta_f, \theta_y, \theta_d) = L_y(\theta_f, \theta_y) - \lambda L_d(\theta_f, \theta_d) \quad (1)$$

$$L_y(\theta_f, \theta_y) = -\frac{1}{n_s} \sum_{i=1}^{n_s} \sum_{k=0}^{K-1} I[y_i^s = k] \log(G_y^k(G_f(x_i^s))) \quad (2)$$

$$L_d(\theta_f, \theta_d) = -\frac{1}{n_s} \sum_{i=1}^{n_s} \log(G_d(G_f(x_i^s))) - \frac{1}{n_t} \sum_{j=1}^{n_t} \log(1 - G_d(G_f(x_j^t))) \quad (3)$$

where θ_f , θ_d , and θ_y are the parameters of G_f , G_d , and G_y , respectively; λ is a regularization coefficient; $G_y^k(\cdot)$ represents the k th element of the output; $I[\cdot]$ is an indicator function; K_s is the number of health states. When the parameters are updated, for L_y : G_f and G_y expect it to be minimized to improve the classification ability of the network; for L_d : G_d expects it to be minimized to enhance the domain discrimination ability and G_f expects it to be maximized to reduce the distribution discrepancy between domains. Therefore, the parameter updates are as follows:

$$(\hat{\theta}_f, \hat{\theta}_y) = \arg \min_{\theta_f, \theta_y} L(\theta_f, \theta_y, \hat{\theta}_d) \quad (4)$$

$$(\hat{\theta}_d) = \arg \max_{\theta_d} L(\hat{\theta}_f, \hat{\theta}_y, \theta_d) \quad (5)$$

where $\hat{\theta}_f$, $\hat{\theta}_y$, and $\hat{\theta}_d$ are the optimal parameters. The above-mentioned updates can be completed by using the stochastic gradient descent (SGD) optimization.

III. PROPOSED METHOD

This section is divided into four parts to introduce our proposed NJTN: construction of the source domain based on simulation data, design of an improved loss function embedded with JMMD, development of a weight assignment mechanism for source-domain samples, and model architecture and optimization algorithm.

A. Construction of Source Domain Based on Simulation Data

In this article, in order to obtain a large number of fault samples with rich label fault information and reduce the dependence on laboratory test rigs, a rotor-bearing system simulation model is built using the lumped-mass method to generate vibration responses of bearings under different states for constructing the source-domain dataset. The dynamic simulation model is shown in Fig. 2. It is mainly composed of a disc, a shaft, a bearing, a base, and other components. The outer ring of the bearing is fixed on the bearing seat, and the inner ring rotates with the shaft. Assuming that the stiffness and damping of the main components are fixed values and the local defects occur at the left support bearing, six differential equations of motion related to the left support bearing can be listed [37]

$$m_{bL}\ddot{x}_{bL} + k_{fLH}(x_{bL} - x_c) + c_{fLH}(\dot{x}_{bL} - \dot{x}_c) + k_{tLH}(x_{bL} - x_{wL}) + c_{tLH}(\dot{x}_{bL} - \dot{x}_{wL}) = 0 \quad (6)$$

$$m_{bL}\ddot{y}_{bL} + k_{fLV}(y_{bL} - y_c) + c_{fLV}(\dot{y}_{bL} - \dot{y}_c) + k_{tLV}(y_{bL} - y_{wL}) + c_{tLV}(\dot{y}_{bL} - \dot{y}_{wL}) = -m_{bL}g \quad (7)$$

$$m_{rL}\ddot{x}_{rL} + k(x_{rL} - x_{rp}) + c_{rb}\dot{x}_{rL} - F_{xbL} = 0 \quad (8)$$

$$m_{rL}\ddot{y}_{rL} + k(y_{rL} - y_{rp}) + c_{rb}\dot{y}_{rL} - F_{ybL} = -m_{rL}g \quad (9)$$

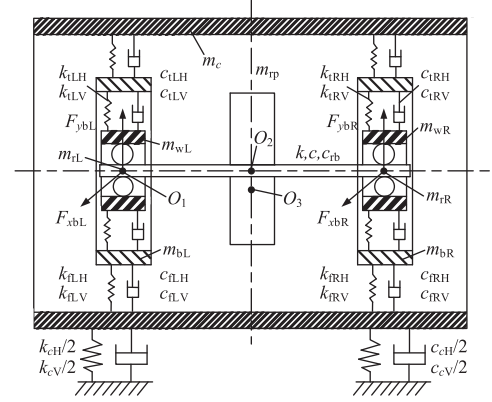


Fig. 2. Schematic diagram of the simulated rotor-bearing system.

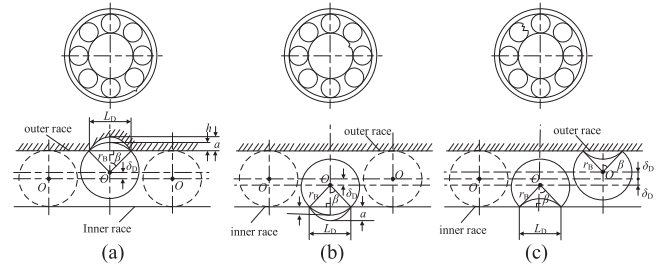


Fig. 3. Schematic of the local defect area. (a) Outer race fault. (b) Inner race fault. (c) Ball fault.

$$m_{wL}\ddot{x}_{wL} + k_{tLH}(x_{wL} - x_{bL}) + c_{tLH}(\dot{x}_{wL} - \dot{x}_{bL}) + F_{xbL} = 0 \quad (10)$$

$$m_{wL}\ddot{y}_{wL} + k_{tLH}(y_{wL} - y_{bL}) + c_{tLH}(\dot{y}_{wL} - \dot{y}_{bL}) + F_{ybL} = -m_{wL}g \quad (11)$$

where m_{bL} is the mass of bearing support; k_{fLH} and K_{fLV} are the lateral and longitudinal support stiffnesses between casing and bearing support, respectively; c_{fLH} and c_{fLV} are the lateral and longitudinal support dampings between casing and bearing support, respectively; k_{tLH} and k_{tLV} are the lateral and longitudinal support stiffnesses between the bearing outer ring and the bearing support, respectively; c_{tLH} and c_{tLV} are the lateral and longitudinal squeeze film dampings between the bearing outer ring and the bearing support; m_{rL} is the equivalent mass of the rotor; k and c_{rb} are the shaft stiffness and the damping of the rotor at bearing, respectively; F_{xbL} and F_{ybL} are the support reaction forces of bearing; and m_{wL} is the mass of bearing outer ring. Specifically, the initial parameters of the rotor-bearing system are consistent with those in the literature [37].

Fig. 3 shows a schematic diagram of bearings with local defect area on the outer ring, inner ring, and ball components. The cyclically varying contact forces F_{xbL} and F_{ybL} are the sum of the contact forces of the balls at different angles

$$F_{xbL} = \sum_{j=1}^Z f_j \cos \theta_j, \quad F_{ybL} = \sum_{j=1}^Z f_j \sin \theta_j \quad (12)$$

where θ_j is the angular position of the j th ball, and f_j is the contact force between the j th ball and the raceway calculated according

TABLE I
MAIN PARAMETERS OF BEARING JIS6306

Descriptions of parameters	Values of parameters
Radius of outer race/mm	63.9
Radius of inner race/mm	40.1
Pitch diameter/mm	52
Diameter of rolling element/mm	11.9
Number of balls	8
Contact angle/ $^{\circ}$	0
Hertzian contact stiffness/ $\text{N} \cdot \text{m}^{-3/2}$	13.34×10^9

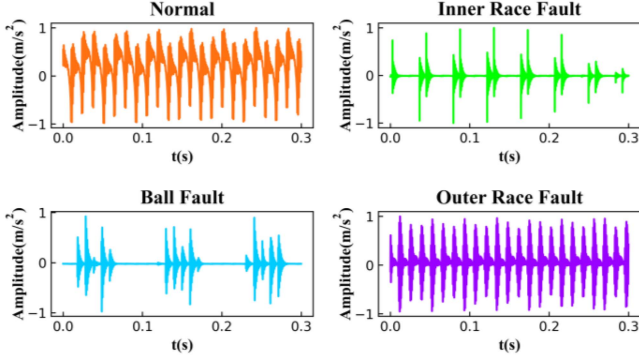


Fig. 4. Time-domain waveforms of simulation data for four states of rolling bearing.

to the following formula:

$$f_j = C_b[\delta_j]^n = C_b(x \cos \theta_j + y \sin \theta_j - \delta_0 - \delta_D)^n \times H(x \cos \theta_j + y \sin \theta_j - \delta_0 - \delta_D) \quad (13)$$

where C_b is the Hertzian contact stiffness, $H(\cdot)$ is the Heaviside function, δ_j is the clearance between the ball and the raceway, δ_0 is the initial clearance, δ_D is the clearance caused by the local defect area, and n is the load-deformation coefficient, which is 2/3 for the ball bearing.

The simulation model adopts a JIS6306 bearing, and its main parameters are listed in Table I. The damage diameter L_D is set to 0.5334 mm, the damage depth a is set to 2.794 mm, and the sampling frequency is set to 10 kHz. Fig. 4 shows a time-domain waveform diagram of the vibration acceleration signal collected by the left support bearing under four different health states when the angular velocity of the shaft is 150 rad/s. The angular velocity of the rotating shaft changes from 150 to 270 rad/s at 5 rad/s intervals, and the horizontal and vertical acceleration signals of the left support bearing are collected to construct the source-domain dataset.

B. Design of Improved Loss Function Embedded With JMMD

Most existing unsupervised domain adaptation methods may misclassify fuzzy samples near the class boundary of the target domain. They ignored the conditional distribution alignment; however, conditional distribution is extremely difficult to characterize as the target domain does not have any labels. Thus, the traditional methods are limited to the marginal distribution alignment of the source and target domains. To address this

problem, this article adopts the JMMD approach [38] to improve the common loss function of the DANN.

The design process of the improved loss function embedded with JMMD is mainly as follows. First, use the output of G_f and G_y in DANN to form the joint distribution of source domain $P_s(G_f(X^s), G_y(G_f(X^s)))$ (denoted as P_s) and the joint distribution of target domain $P_t(G_f(X^t), G_y(G_f(X^t)))$ (denoted as P_t), where $X^s = \{x_i^s\}_{i=1}^{n_s}$ and $X^t = \{x_i^t\}_{i=1}^{n_t}$. Second, calculate the discrepancy between P_s and P_t by reproducing kernel Hilbert space embeddings, denoted as $D(P_s, P_t)$. Finally, use the squared distance between kernel mean embeddings to compute the empirical estimate of $D(P_s, P_t)$, denoted as $\hat{D}(\theta_f, \theta_y)$. Let $\hat{f}_i^s = G_f(x_i^s)$, $\hat{f}_i^t = G_f(x_i^t)$, $\hat{y}_i^s = G_y(G_f(x_i^s))$, and $\hat{y}_i^t = G_y(G_f(x_i^t))$, then the following JMMD formula can be obtained:

$$\begin{aligned} \hat{D}(\theta_f, \theta_y) = & \frac{1}{n_s^2} \sum_{i=1}^{n_s} \sum_{j=1}^{n_s} k(\hat{f}_i^s, \hat{f}_j^s) \cdot k(\hat{y}_i^s, \hat{y}_j^s) \\ & + \frac{1}{n_t^2} \sum_{i=1}^{n_t} \sum_{j=1}^{n_t} k(\hat{f}_i^t, \hat{f}_j^t) \cdot k(\hat{y}_i^t, \hat{y}_j^t) \\ & - \frac{1}{n_s n_t} \sum_{i=1}^{n_s} \sum_{j=1}^{n_t} k(\hat{f}_i^s, \hat{f}_j^t) \cdot k(\hat{y}_i^s, \hat{y}_j^t) \end{aligned} \quad (14)$$

where $k(\cdot)$ is the kernel function, and considering the generality of Gaussian kernel, we choose Gaussian kernel function with bandwidth set to median pairwise squared distances on the training data [38].

Despite the fact that (14) captures the interaction between different variables in joint distributions P_s and P_t and explicitly calculates the joint distribution discrepancy between the source domain and the target domains, it is usually still not enough to accurately characterize the dynamically changing data distribution. In contrast, the discrepancy learned by DANN, which is an implicit distance, can be learned dynamically in the data. Therefore, through embedding (14) into (1) (i.e., the common loss function of the DANN) for improvement, on one hand, the above-mentioned problem can be alleviated to a certain extent; on the other hand, using JMMD to align the joint distributions of source and target domains and reducing their marginal distribution discrepancy through DANN can simultaneously help adapt their conditional distributions in the unsupervised scenario, which can overcome the limitations on separate marginal distribution alignment.

After that, the proposed improved loss function embedded with JMMD is

$$L(\theta_f, \theta_y, \theta_d) = L_y(\theta_f, \theta_y) - \lambda L_d(\theta_f, \theta_d) + \mu \hat{D}(\theta_f, \theta_y) \quad (15)$$

with

$$\lambda, \mu \sim 2m/(1 + e^{-Kt}) - m \quad (16)$$

where μ is also a regularization coefficient, and the growth trajectory of regularization coefficients λ and μ is given in (16), in which t is the current iterative training time, m is the upper

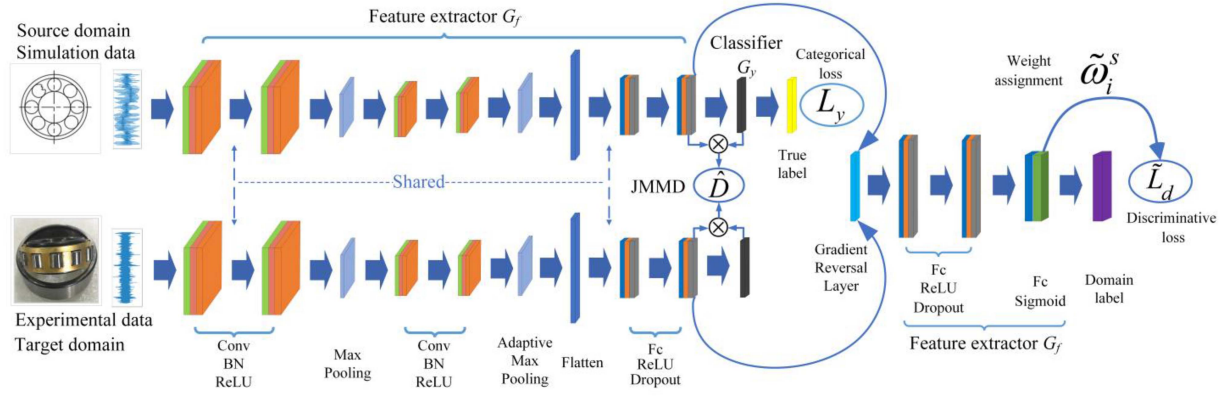


Fig. 5. Model architecture of the proposed method.

limit of regularization coefficient, and K controls the rate of change.

C. Development of Weight Assignment Mechanism for Source-Domain Samples

The classic unsupervised domain adversarial adaptation methods usually assign equal weight to each source-domain sample to adapt to the target domain. However, there may be samples in the source domain that are excessively different from the target domain. At this time, if the equal weight allocation strategy is still used, it is likely to cause a negative transfer of the diagnostic model [39].

In this article, in order to characterize the similarity of samples in different domains to suppress negative transfer, a weight assignment mechanism for source-domain samples is developed. Considering that the samples with higher similarity in different domains are more difficult to separate whereas samples with larger discrepancies are easier to distinguish for the domain discriminator, the specific weights assigned to each source-domain sample can be calculated according to the domain prediction error of the domain discriminator, defined as follows:

$$\omega_i^s = -\log(G_d(G_f(x_i^s))) \quad (17)$$

where ω_i^s denotes the specific weight of the i th sample in the source domain, and after using min—max normalization, the normalized weight of ω_i^s is given as follows:

$$\tilde{\omega}_i^s = \frac{\omega_i^s - \omega_{i,\min}^s}{\omega_{i,\max}^s - \omega_{i,\min}^s} \quad (18)$$

where $\omega_{i,\max}^s = \max(\omega_i^s)$ and $\omega_{i,\min}^s = \min(\omega_i^s)$. The normalized weights are introduced into (3), and then the new domain discriminative cross-entropy loss can be expressed by

$$\begin{aligned} \tilde{L}_d(\theta_f, \theta_d) = & -\frac{1}{n_s} \sum_i \tilde{\omega}_i^s \log(G_d(G_f(x_i^s))) \\ & -\frac{1}{n_t} \sum_j \log(1 - G_d(G_f(x_j^t))). \end{aligned} \quad (19)$$

The final improved loss function can be obtained by combining (19) and (15), defined as follows:

$$L(\theta_f, \theta_y, \theta_d) = L_y(\theta_f, \theta_y) - \lambda \tilde{L}_d(\theta_f, \theta_d) + \mu \hat{D}(\theta_f, \theta_y). \quad (20)$$

D. Model Architecture and Optimization Algorithm

Fig. 5 shows the model architecture of the proposed method, which is mainly composed of three modules: feature extractor G_f , classifier G_y , and domain discriminator G_d . Specifically, G_f consists of four one-dimensional (1-D) convolution (Conv) layers, two Max pooling layers, and two fully-connected (Fc) layers. Each Conv layer comes with a 1-D batch normalization layer and a rectified linear unit (ReLU) activation function, and each Fc layer comes with a ReLU activation function and a dropout layer; G_y consists of an Fc layer; G_d consists of three Fc layers, two of which come with ReLU activation functions and dropout layers, and the remaining one comes with a Sigmoid activation function.

The proposed model is trained using the SGD algorithm, and the parameters θ_f , θ_d , and θ_y will be updated as follows:

$$\begin{aligned} \theta_f & \leftarrow \theta_f - \varepsilon(\partial L_y / \partial \theta_f - \lambda \partial \tilde{L}_d / \partial \theta_f + \mu \partial \hat{D} / \partial \theta_f) \\ \theta_y & \leftarrow \theta_y - \varepsilon(\partial L_y / \partial \theta_y + \mu \partial \hat{D} / \partial \theta_y) \\ \theta_d & \leftarrow \theta_d - \varepsilon \lambda \partial \tilde{L}_d / \partial \theta_d \end{aligned} \quad (21)$$

where ε is the learning rate.

In each epoch, after the training phase of the model, the target-domain test set will enter the model and they will only go through G_f and G_y ; then G_y will give the classification results. The specific training and test procedures of the proposed model are shown in Algorithm 1.

IV. EXPERIMENT AND RESULT ANALYSIS

A. Introduction of Target-Domain Datasets and Comparison Methods

The effectiveness of the proposed method is verified in two cases of unsupervised cross-domain bearing fault diagnosis, in which *Case 1* uses dataset from the QPZZ-II bearing fault

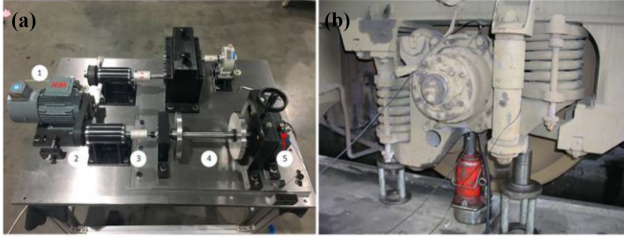


Fig. 6. Bearings installed in different experimental devices. (a) QPZZ-II bearing fault test rig. (b) Electric locomotive bearing fault test rig.

Algorithm 1: Training and Test Procedures of the Proposed Model.

Input: The preprocessed source-domain labeled dataset

$\mathcal{D}_s = \{x_i^s, y_i^s\}_{i=1}^{n_s}$ and the preprocessed target-domain unlabeled dataset $\mathcal{D}_t = \{x_i^t\}_{i=1}^{n_t}$

- 1: Set model hyperparameters such as convolutional layer, pooling layer, activation function, learning rate, iteration number, and batch size
- 2: Randomly initialize the weights and biases of the model
- 3: Divide the training and test sets for the source and target domains respectively
- 4: **For** each training epoch **do**
- 5: **If** phase = train **then**
- 6: **For** each data batch **do**
- 7: Calculate the output of G_f and G_y
- 8: Solve $\hat{D}(\theta_f, \theta_y)$ based on Eq. (14)
- 9: Solve $L_y(\theta_f, \theta_y)$ based on Eq. (1)
- 10: Calculate the output of G_f and G_d
- 11: Solve $\tilde{\omega}_i^s$ based on Eq. (17) and Eq. (18)
- 12: Solve $\tilde{L}_d(\theta_f, \theta_d)$ based on Eq. (19)
- 13: Update model parameters $\theta_f, \theta_d, \theta_y$ based on Eq. (21)
- 14: **End For**
- 15: **End If**
- 16: **If** phase = test **then**
- 17: Calculate the output of G_y
- 18: **End If**
- 19: **End For**

Output: Optimized model parameters $\theta_f, \theta_d, \theta_y$ and classification results on the test set

test rig [40] to construct Target domain 1, and *Case 2* uses dataset from the electric locomotive bearing fault test rig [41] to construct Target domain 2. Figs. 6 and 7, respectively, show the experimental devices and three kinds of fault bearings in two cases. The sampling frequencies of the two cases are 25.6 and 12.8 kHz, respectively. In the two cases, the sample dimension is 1000, and the sample setting of the source domain is consistent with the selected target domain. The time-domain waveforms of target-domain bearing under four health states in two cases are shown in Figs. 8 and 9, respectively. The details of the two target-domain datasets are listed in Table II. We perform de-mean, “0-1” normalized data processing for all data points under each fault class in the source and target domains before the samples are fed into the model, respectively.



Fig. 7. Three kinds of fault bearings in different cases. (a) Case 1; (b) Case 2.

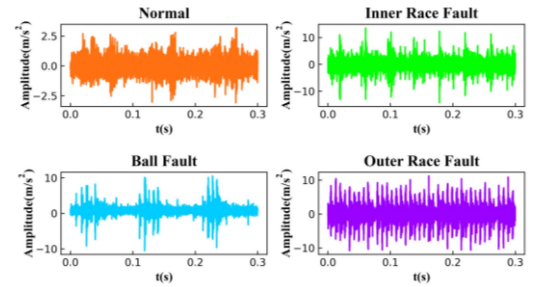


Fig. 8. Time-domain waveforms of rolling bearing under four states in Case 1.

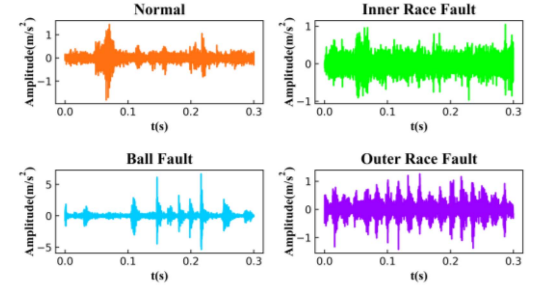


Fig. 9. Time-domain waveforms of rolling bearing under four states in Case 2.

TABLE II
DETAILS OF TARGET-DOMAIN DATASETS

Datasets	Bearing types	Health states	Number of training/test samples	Motor speeds
Target domain 1 (Case 1)	NU205EM	Normal Inner race fault Ball fault Outer race fault	320/80	1500 rpm
Target domain 2 (Case 2)	552732QT	Normal Inner race fault Ball fault Outer race fault	480/120	490 rpm 498 rpm 531 rpm 490 rpm

In order to verify the superiority of the proposed method, it is compared with the other seven methods.

- 1) First, the proposed method (denoted as *Method 1*) is compared with the standard convolutional neural network (denoted as *Method 2*) to verify the necessity of a transfer diagnosis strategy.

TABLE III

COMPARISONS OF UNSUPERVISED CROSS-DOMAIN DIAGNOSIS RESULTS OF THE EIGHT METHODS IN TWO CASES

Methods	Case 1 (%)	Case 2 (%)
Method 1	85.532±1.359	81.355±1.880
Method 2	25.812±0.648	24.250±2.284
Method 3	79.312±3.103	61.189±6.846
Method 4	81.281±2.688	73.334±14.556
Method 5	79.937±4.219	67.771±12.536
Method 6	47.094±10.120	39.270±11.399
Method 7	82.688±2.283	77.145±2.641
Method 8	82.156±2.138	75.645±2.092

The bold entities highlight the proposed method.

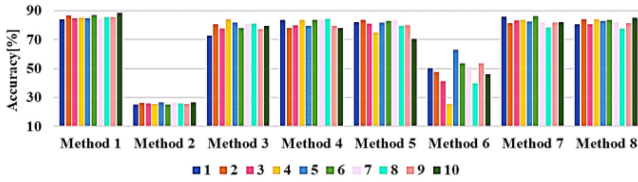


Fig. 10. Unsupervised cross-domain diagnosis accuracies of the eight methods during the ten repetitions in Case 1.

- 2) Secondly, for reflecting the advantages of the proposed method, it is compared with some popular unsupervised cross-domain fault diagnosis methods, including DANN, deep adaptation network, joint adaptation network, and CORAL, which are successively denoted as *Method 3* to *Method 6*.
- 3) Finally, to respectively illustrate the effectiveness of the designed improved loss function embedded with JMMD and the developed weight allocation mechanism for source-domain samples, DANN combined with only the improved loss function, and only the weight allocation mechanism are used for comparisons, which are denoted as *Method 7* and *Method 8*.

Ten trials for each method are run in each case to reduce the random influence. In addition, based on the idea of a greedy algorithm, we experimentally choose the value of a variable under the premise that other hyperparameter variables are determined in order to find a relatively better combination of hyperparameters. In the two cases, the main parameters of the proposed method are set as follows: The learning rate is 0.0001, the batch size is 32, the iteration number N is 300, the upper limit of regularization coefficient m is 1, the rate of change K is 1/30, the value of dropout probability is 0.5 (default value), the optimizer is Adam (the values of first-order moment decay coefficient and second-order moment decay coefficient are 0.9 and 0.999, respectively), and the weight decay is 1×10^{-5} .

B. Analysis of the Comparison Results

In this article, the average accuracy, average F1-score, and the confusion matrix are used to fully compare the diagnostic performance of each method on the test set of the target domain. The results are shown in Table III and Figs. 10–14, respectively. In the confusion matrix, the labels of normal, inner race fault, ball fault, and outer race fault states are set as 0, 1, 2, and 3,

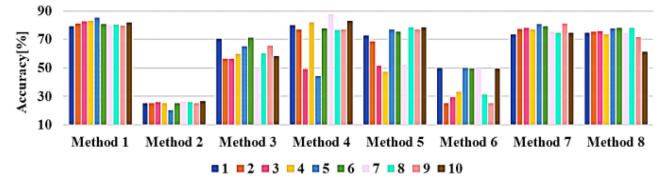


Fig. 11. Unsupervised cross-domain diagnosis accuracies of the eight methods during the ten repetitions in Case 2.

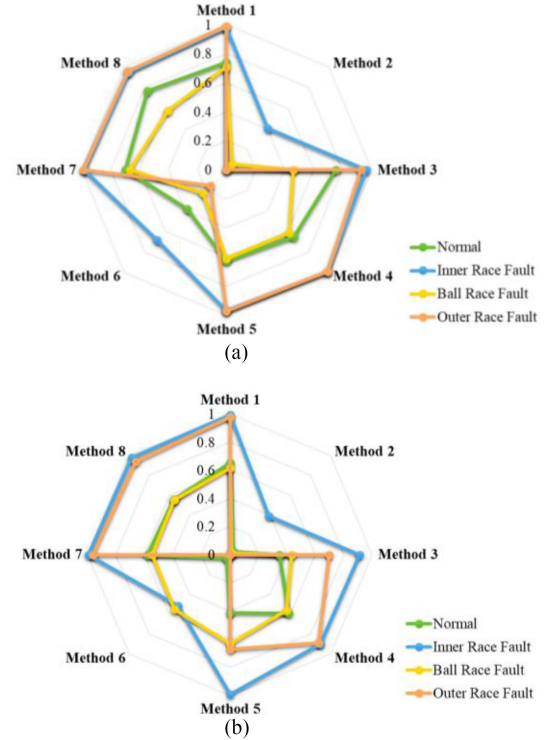


Fig. 12. Average F1-scores of four states of each method in different cases. (a) Case 1. (b) Case 2.

respectively; the abscissa is the predicted label, and the ordinate is the real label.

The following main conclusions can be drawn:

- 1) Through the comparisons between *Method 1* and *Methods 2–6*, it can be seen that *Method 1* has achieved the highest fault diagnosis accuracies. When faced with the unsupervised scenario, the features learned by the proposed method are more representative and easier to distinguish. Specifically, in the two cases, the average diagnostic accuracies of *Method 1* are 85.532 and 81.355%, respectively; however, the highest accuracies of other methods are only 81.281 and 73.334%, respectively.
- 2) The designed improved loss function embedded with JMMD helps enhance the diagnosis accuracy of each fault category, especially the ball fault and outer race fault states, which can be found by the comparisons between *Method 7* and *Method 3*, as well as *Method 1* and *Method 8*. Concretely, in Case 1, the F1-scores of *Method 7* are 0.664 and 0.987, respectively, whereas those of *Method 3* are successively 0.458 and 0.928; the F1-scores of

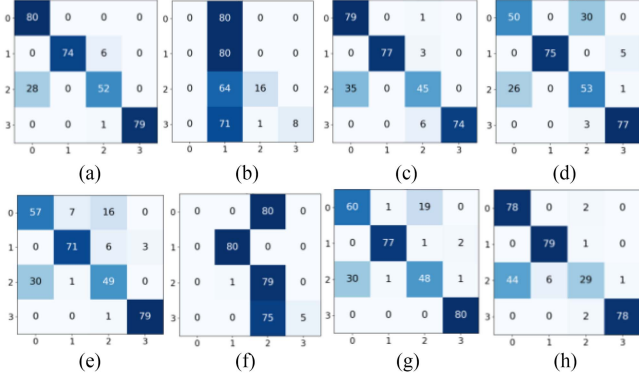


Fig. 13. Confusion matrices for the third trial of each method in Case 1. (a) Method 1. (b) Method 2. (c) Method 3. (d) Method 4. (e) Method 5. (f) Method 6. (g) Method 7. (h) Method 8.

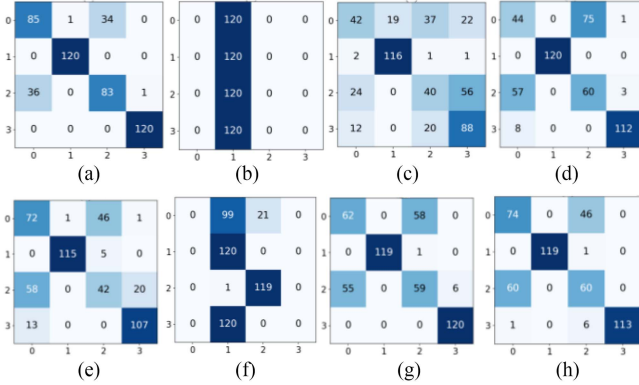


Fig. 14. Confusion matrices for the third trial of each method in Case 2. (a) Method 1. (b) Method 2. (c) Method 3. (d) Method 4. (e) Method 5. (f) Method 6. (g) Method 7. (h) Method 8.

Method 1 are 0.708 and 0.989, respectively; however, *Method 8* has those of 0.574 and 0.953, accordingly. In *Case 2*, the F1-scores of *Method 7* are 0.545 and 0.962, respectively, whereas those of *Method 3* are 0.434 and 0.697, respectively; the F1-scores of *Method 1* are 0.616 and 0.980, respectively, whereas those of *Method 8* are 0.530 and 0.904, respectively.

- 3) It is not difficult to find out that the developed weight allocation mechanism for source-domain samples plays a positive role in improving the diagnostic accuracy and stability through the comparisons between *Method 8* and *Method 3*, as well as *Method 1* and *Method 7*. In spite of the fact that adapting the samples in the source domain that are excessively different from the target domain can cause negative transfer, the mechanism can adaptively assign different weights for each source-domain sample by measuring their similarities with the target-domain samples to suppress negative transfer to a great degree.

In order to show the performance of feature adaptation of the proposed method, *t*-distributed stochastic neighbor embedding (*t*-SNE) is used to reduce the dimension of the output feature

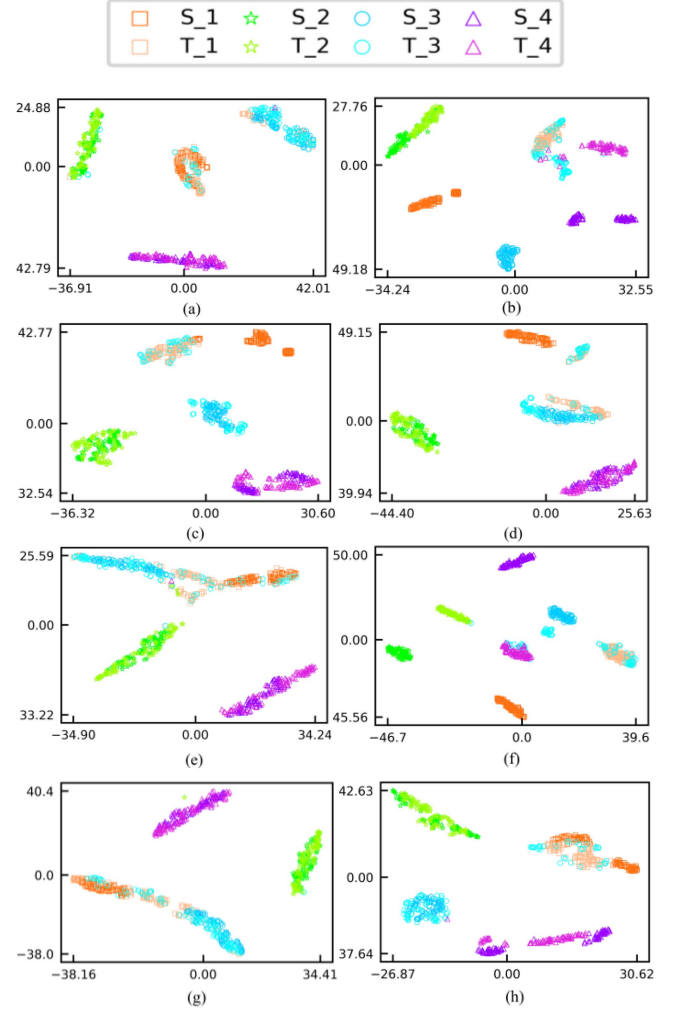


Fig. 15. Visual feature adaptation results for each method using *t*-SNE in Case 1. (a) Method 1. (b) Method 2. (c) Method 3. (d) Method 4. (e) Method 5. (f) Method 6. (g) Method 7. (h) Method 8.

vector to achieve visual results. For the test sets of the source domain and target domain, the feature adaptation results of the third trial of each method in the two cases are shown in Figs. 15 and 16, respectively, in which the labels of normal, inner race fault, ball fault, and outer race fault states in the source domain are set as S_0, S_1, S_2, and S_3, and the target-domain ones are T_0, T_1, T_2, and T_3. The abscissa is feature component 1, and the ordinate is feature component 2.

According to the comparison results of feature adaptation, the deep features of the source and target domains extracted by the proposed method are more similar than others, i.e., they have better domain-invariant properties. It is because the proposed method takes the adaptations of marginal distribution and conditional distribution into account, which significantly reduces the distribution discrepancy between domains. At the same time, the proposed method can adaptively give the specific weights for each source-domain sample, so as to selectively transfer the diagnosis knowledge in the source domain to suppress negative transfer.

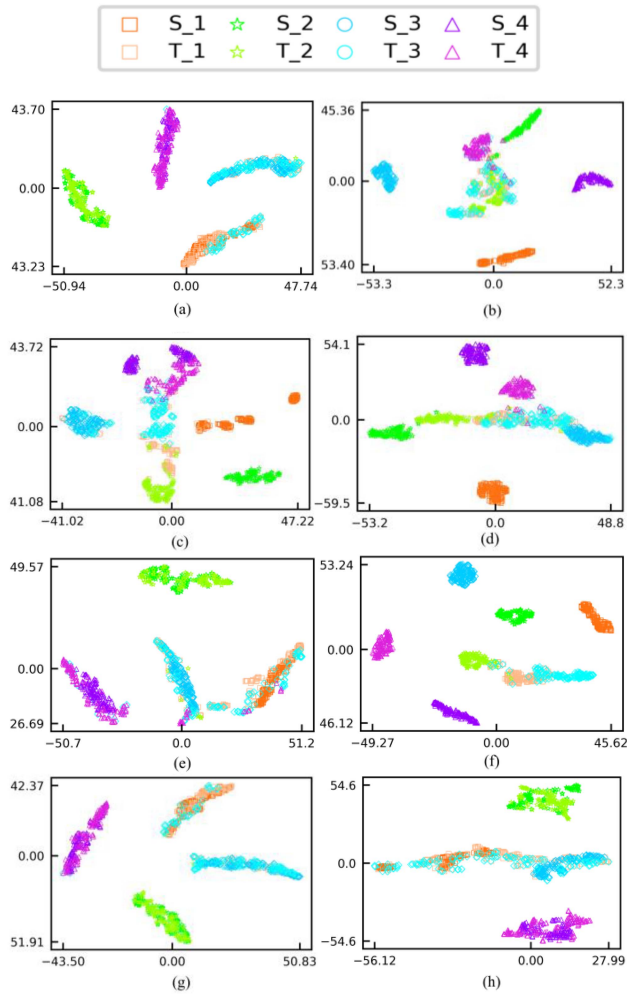


Fig. 16. Visual feature adaptation results for each method using t -SNE in Case 2. (a) Method 1. (b) Method 2. (c) Method 3. (d) Method 4. (e) Method 5. (f) Method 6. (g) Method 7. (h) Method 8.

V. CONCLUSION

This article presented a novel method called the NJTN to further the unsupervised cross-domain diagnostic performance in bearing fault detection. In the proposed method, the source domain was constructed by the bearing simulation data containing rich fault label information to reduce the dependence on the resources of laboratory test rigs. The alignments of marginal and conditional distributions were fully considered by the designed improved loss function embedded with JMMD in unsupervised scenarios, which significantly reduces the distribution discrepancy between domains. The developed weight allocation mechanism for source-domain samples was used to suppress negative transfer.

The effectiveness and superiority of the proposed method were demonstrated by two transfer fault diagnosis cases of bearing from the simulation domain to the experimental domain in unsupervised scenarios. Our future work will focus on further improving the generalization performance of the model: On one hand, considering that the label sets of source and target domains are hard to be consistent in real application, we will explore how

to complete transfer fault diagnosis tasks in label-inconsistent scenarios. On the other hand, considering that a single source domain might not be enough for a classifier to learn sufficient information about the fault features, we will examine how to transfer the fault diagnosis knowledge of multiple source domains to the target domain.

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